

American University of Armenia

CS337 Cybersecurity

Email Phishing Detection and Classification

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Abstract

This project focuses on the detection of phishing emails using machine learning techniques. With the increasing number of cyber threats and fraudulent communication, identifying malicious emails has become a critical task for ensuring digital security. The objective of this study is to develop and evaluate classification models capable of distinguishing between legitimate and phishing emails based on their content.

Several machine learning algorithms were applied, including Naive Bayes, Logistic Regression and Random Forest, to analyse and classify email data. The dataset was pre-processed and explored through data analysis and visualization techniques to better understand feature distributions and patterns. The models were trained and evaluated using standard performance metrics to assess their effectiveness.

The results demonstrate that machine learning approaches can successfully identify phishing emails with a high level of accuracy. Additionally, the project explores the practical application of the trained model by proposing a chatbot-like interface, where users can input email content and receive real-time classification results.

This study highlights the potential of machine learning in enhancing cybersecurity solutions and provides a foundation for further improvements, such as integrating more advanced models and real-world deployment scenarios.

The full implementation, including the Jupyter Notebook (.ipynb) source code, is available in the project's [GitHub repository](#).

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Literature Review

Before the implementation phase, a literature review was conducted to better understand existing approaches to phishing detection and to identify effective techniques used in previous research.

The first paper, [*“Improved Phishing Detection using Model-Based Features”*](#) focuses on enhancing phishing detection by using carefully engineered features extracted from email content and metadata. The study demonstrates that combining multiple types of features can significantly improve classification accuracy. This influenced the feature extraction phase of this project, where both textual and structural characteristics of emails were considered.

The second paper, [*“A Survey of Phishing Email Filtering Techniques”*](#) provides a comprehensive overview of different methods used for phishing detection, including rule-based systems, machine learning approaches, and hybrid techniques. The survey highlights the strengths and limitations of each method, emphasizing that machine learning models generally offer better scalability and adaptability.

The third paper, [*“Study on Phishing Attacks”*](#) analyzes common phishing strategies and attacker behavior. It outlines typical characteristics of phishing emails, such as urgent language, deceptive links, and impersonation of trusted entities. These insights were directly applied during the phishing analysis stage to define detection rules and identify relevant features.

Overall, the literature review provided a solid theoretical foundation for the project, helping to justify the chosen methodology and ensuring that the implemented solution aligns with established research in the field of cybersecurity.

Problem Statement

Phishing emails continue to be one of the most widespread and dangerous cybersecurity threats in the digital environment. Attackers increasingly use deceptive techniques to trick users into revealing sensitive information such as passwords, banking credentials, and personal data. These

emails often mimic legitimate communication, making them difficult to distinguish from genuine messages.

A significant challenge lies in the fact that many users lack the awareness or technical expertise required to accurately identify phishing attempts. As a result, malicious emails frequently go unnoticed, leading to financial losses, data breaches, and compromised systems. Traditional manual detection methods are not sufficient to handle the growing volume and sophistication of such attacks.

Therefore, there is a clear need for an automated system capable of detecting phishing emails efficiently and accurately. Such a system should be able to analyze email content, identify suspicious patterns, and classify messages with minimal human intervention. Addressing this problem can significantly enhance user security and reduce the risks associated with phishing attacks.

Methodology

Dataset Selection and Collection

The first step of the project was the selection of an appropriate [dataset](#) for phishing email detection. For this purpose, publicly available email datasets were used, containing both phishing and legitimate emails. The inclusion of both classes was essential to train a supervised machine learning model capable of distinguishing between malicious and non-malicious messages.

The dataset was chosen based on its relevance, availability, and the presence of labeled data, which is required for classification tasks. Using real-world email examples allowed the model to learn patterns that are commonly associated with phishing attacks.

Feature	Description
sender	The email address of the sender. It helps identify the origin and detect suspicious domains.
receiver	The email address of the recipient. Useful for contextual analysis of communication.
date	The timestamp of when the email was sent. Can help detect unusual sending patterns.
subject	The subject line of the email, often containing misleading or urgent phrases.
body	The main content of the email, where phishing indicators such as requests for sensitive data appear.
urls	Links included in the email, which may point to malicious or deceptive websites.
label	Indicates whether the email is phishing (1) or legitimate (0).

Fig.1 Dataset Overview

Data Understanding and Exploration

After selecting the dataset, an initial data exploration phase was conducted. This step aimed to better understand the structure and characteristics of the data.

The emails were analyzed to identify common phishing indicators such as:

1. the use of suspicious or urgent language,
2. the presence of malicious or shortened links,
3. mismatches in sender information,
4. and requests for sensitive data.

Basic visualizations and statistical analysis were used to examine the distribution of phishing versus legitimate emails. This helped ensure that the dataset was suitable for training and did not contain major imbalances or inconsistencies that could affect model performance.

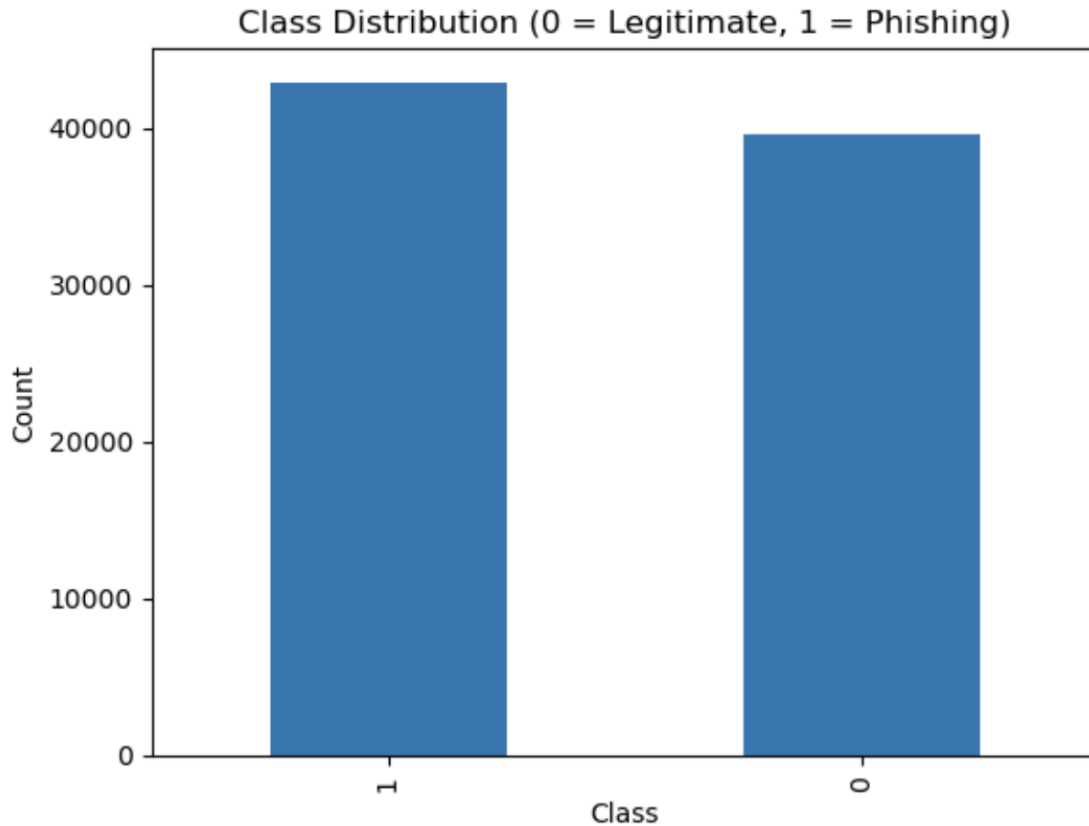


Fig.2 Class Distribution

Data Pre-processing

Before applying any models, the dataset underwent preprocessing to convert raw email text into a format suitable for analysis. This included:

1. cleaning the text (removing unnecessary symbols, punctuation, and noise),
2. normalizing the text (such as converting to lowercase),
3. and preparing the data for feature extraction.

Pre-processing is a critical step, as machine learning models cannot directly work with raw text and require structured numerical input.

Feature Extraction

To enable machine learning classification, email content was transformed into meaningful features. Several types of features were considered, including:

1. the number of links in the email,
2. the presence of urgent or suspicious keywords,
3. characteristics of the sender,
4. and general text-based representations.

These features were used to capture patterns that differentiate phishing emails from legitimate ones. The transformation of text into numerical vectors allowed the models to process and learn from the data effectively.

Machine Learning Model Training

Several machine learning algorithms were applied to classify emails, including: Naive Bayes, Logistic Regression and Random Forest.

These models were trained on the processed dataset to learn patterns associated with phishing behavior. Each model has different strengths:

1. Naive Bayes is efficient and performs well on text data,
2. Logistic Regression provides interpretable results,
3. Random Forest captures complex relationships between features.

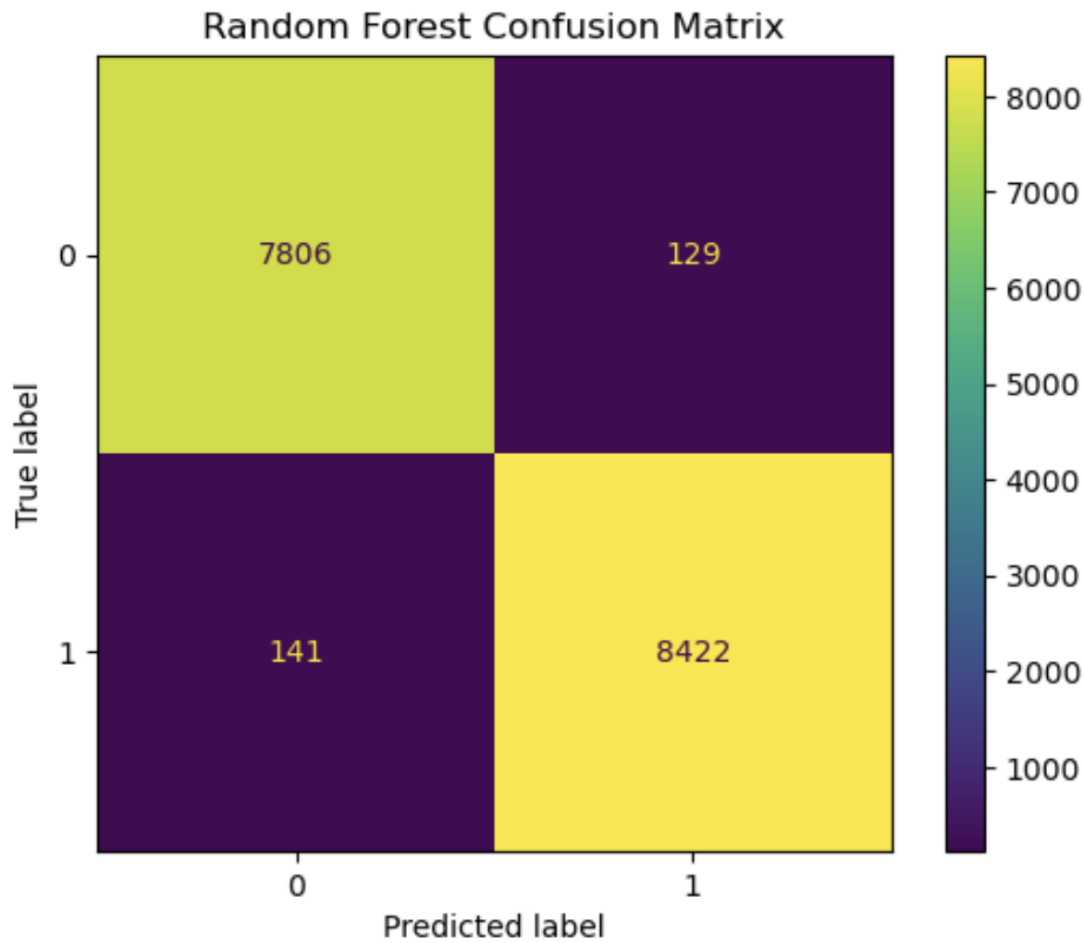


Fig.3 Random Forest Confusion Matrix

Model Evaluation

To assess the performance of the models, standard evaluation metrics were used, including:

1. accuracy,
2. precision,
3. recall,
4. and F1-score.

These metrics provided a comprehensive understanding of how well the models performed, especially in detecting phishing emails without misclassifying legitimate ones.

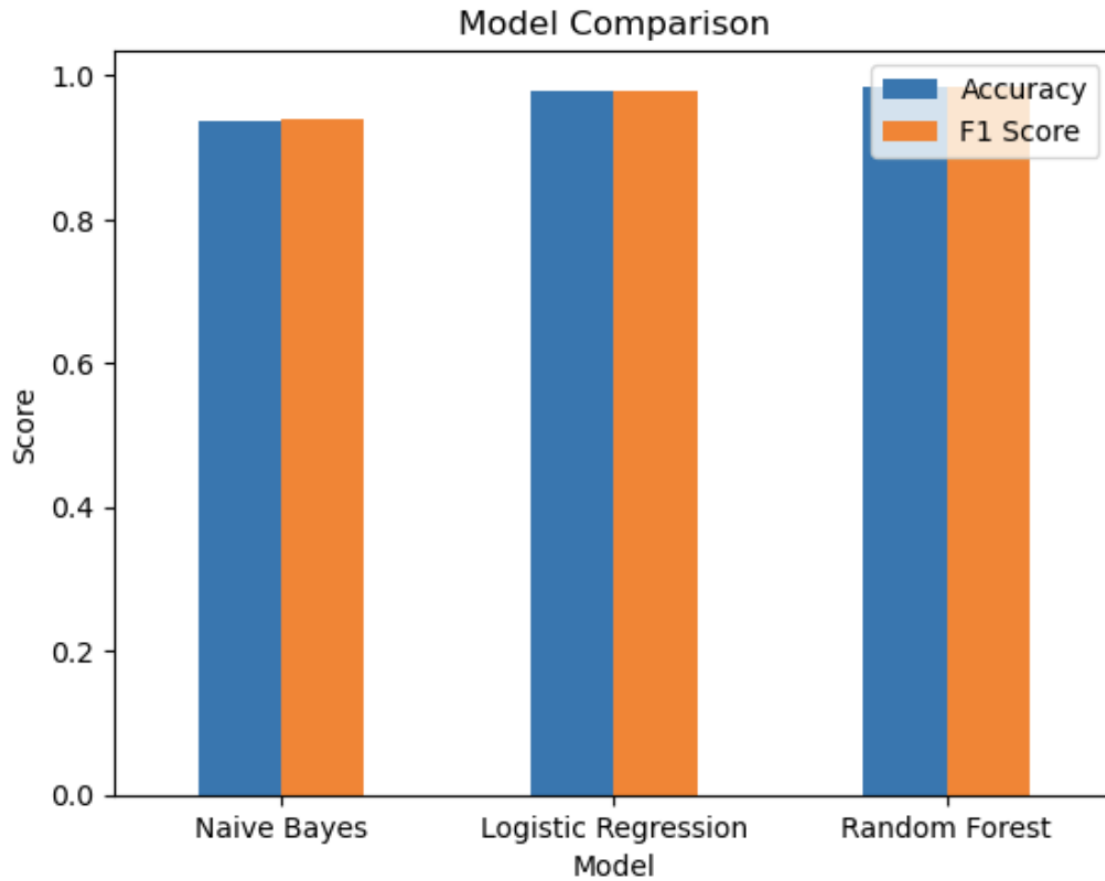


Fig.4 Model Comparison

System Implementation Concept

Finally, the trained model was considered for integration into a simple application interface. The idea was to simulate a chatbot-like system where a user can input an email and receive a classification result (phishing, or legitimate).

This step demonstrates the practical applicability of the developed solution and highlights its potential use in real-world scenarios.

Conclusion

In this project, an automated system for phishing email detection was successfully developed using a machine learning approach. The study began with the analysis of phishing characteristics and

continued through dataset selection, preprocessing, feature extraction, and model training. By applying different classification algorithms, including Naive Bayes, Logistic Regression, and Random Forest, the system was able to effectively distinguish between phishing and legitimate emails. The results demonstrate that machine learning models provide reliable and scalable solutions for detecting phishing attempts.

Overall, this project highlights the importance of combining data-driven techniques with domain knowledge in cybersecurity tasks. The developed solution not only achieves accurate classification but also shows potential for real-world application through a user-oriented interface. Future improvements may include the use of larger and more diverse datasets, advanced natural language processing techniques, and deployment as a fully functional application.